

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	28 February 2026
Team ID	LTVIP2026TMIDS88469
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Technical Architecture:

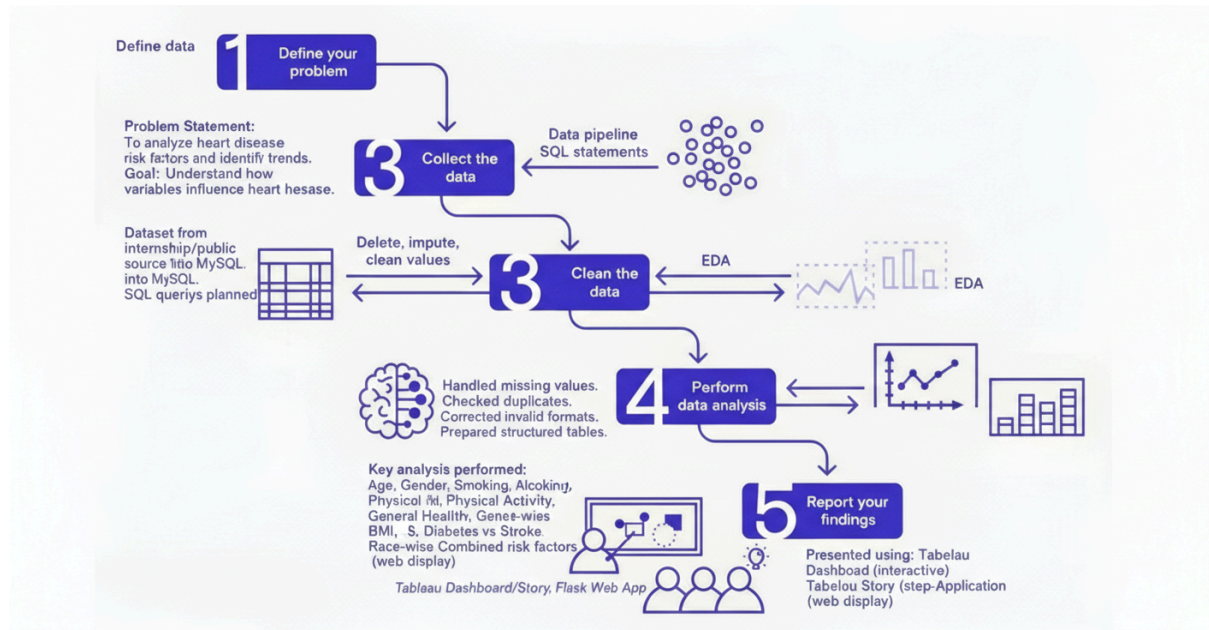


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for viewing dashboards and insights	HTML, CSS, JavaScript, Tableau Public Embedding
0.	Data Processing Logic	Data cleaning & preprocessing scripts	Python (Pandas, NumPy)
0.	Data Storage	Stores raw data and cleaned datasets	CSV files + MySQL Database
0.	Visualization Layer	Creates interactive visual dashboards and charts	Tableau Public / Tableau Desktop
0.	Infrastructure (Server / Hosting)	Hosts any scripts and serves embedded dashboards	Local Machine or Cloud Hosting (Render / Railway)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Uses open-source Python libraries for data processing	Python (Pandas, NumPy), Flask
2	Security	Data is stored securely in MySQL and dashboards shared via controlled links	MySQL authentication, Tableau Public permissions
3	Scalable Architecture	3-tier architecture: UI → Backend → Database	Flask + MySQL + Tableau
4	Availability	Dashboards accessible anytime through Tableau Public and Flask deployed link	Tableau Public, Render / Railway
5	Performance	Small dataset loads fast; Tableau extracts improve dashboard speed	Tableau Data Extracts, Python ETL